

Variables

Variables and Constants

- Variables – Vary from person to person or object to object.
- A **variable** is a characteristic or condition that can change or take on different values.
- Constants – Remain constant from person to person or object to object.

Example

A study is conducted to determine if there are gender differences in fine motor skills among five year olds from middle class families.

What are the variables in the study? Gender, Fine motor skills

- *What are the constants in the study?* Age, Socio-economic status

Note: Whether a trait is a constant or a variable is determined by the nature of the particular study.

Types of Variables

- Quantitative:
 - Numeric value that it makes sense to do arithmetic operations (+, -, x, /)
 - Finding an arithmetic “average” makes sense
 - EX:
 - Height
 - Weight
 - Age
 - Income
 - Test Scores

Types of Variables

- Categorical
 - Records which of several groups or categories to which an individual belongs
 - If there are only 2 possible categories, the variable is called binary
 - Ex:
 - Gender
 - Race
 - SES (socio-economic status)
 - Marital status

Caution!!!

- Most quantitative variables can be asked in such a way as to make them a categorical variable.
 - Ex: Age:
 - Please choose the category that describes how old you are
 - 0-19 20-39 40-59 60-79 OVER 80

Determine if the variable is Quantitative or Categorical

- Height
- Armspan
- Whether or not the individual went to sleep before 12:00am
- Month of birth
- Distance from home
- Whether the individual has a cell phone
- How many e-mail messages a person has sent in the last 24 hours
- The age of a bride on her wedding day
- Whether the bride is older, younger, or the same age as the groom in a wedding couple
- The number of letters in a person's last name

Check your answers on the next slide



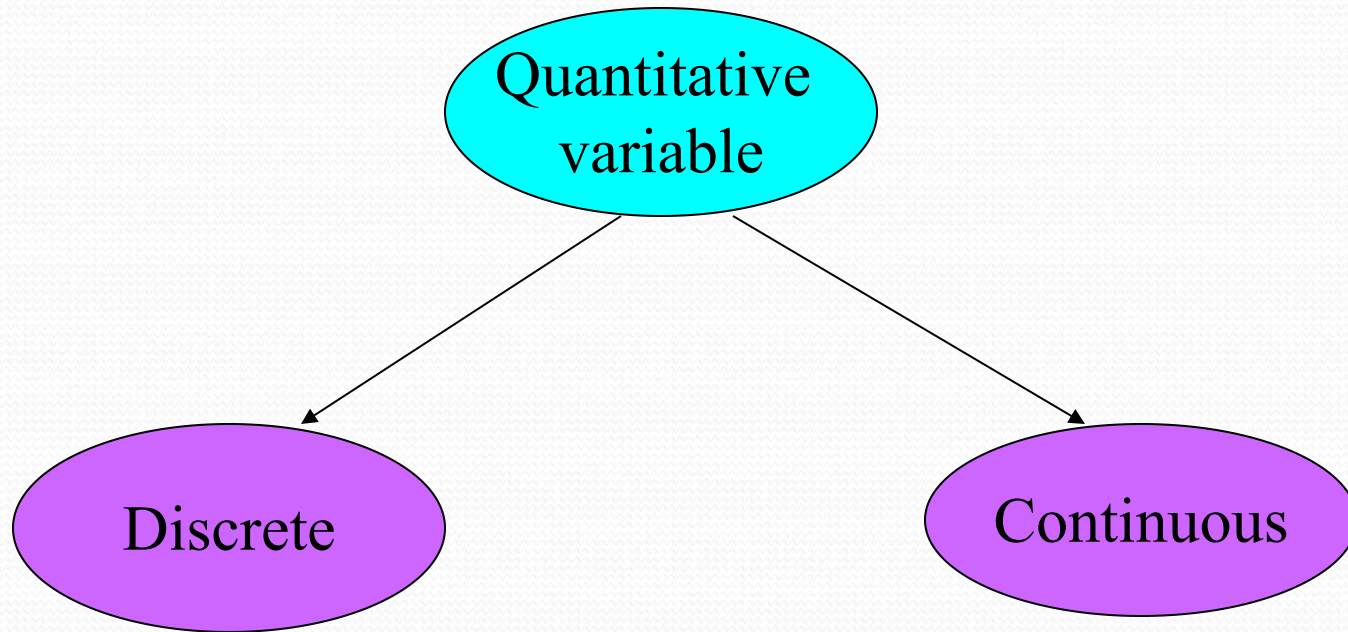
➤ Quantitative Variables

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Categorical

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Quantitative variable can be classified as
'Discrete or Continuous'



Types of Variables

- **Discrete variables** (such as class size) consist of indivisible categories.
- **Continuous variables** (such as time or weight) are infinitely divisible into whatever units a researcher may choose. For example, time can be measured to the nearest minute, second, half-second, etc.

Discrete and Continuous Variables

- A variable is **discrete** if the values it takes on are integers or can be thought of in some unit of measurement in which they are integers.
- A variable is **continuous** if in any unit of measurement, whenever it can take on the values a and b , it can also theoretically take on all the values between a and b .

Examples

Classify each of the following variables as discrete or continuous:

- Number of earned credits to date Discrete
- Height Continuous
- Family Income Discrete
- Number of siblings Discrete

Qualitative and quantitative variables may be further subdivided:

